

Customer Churn Analysis Report

1. Visualizations and Insights

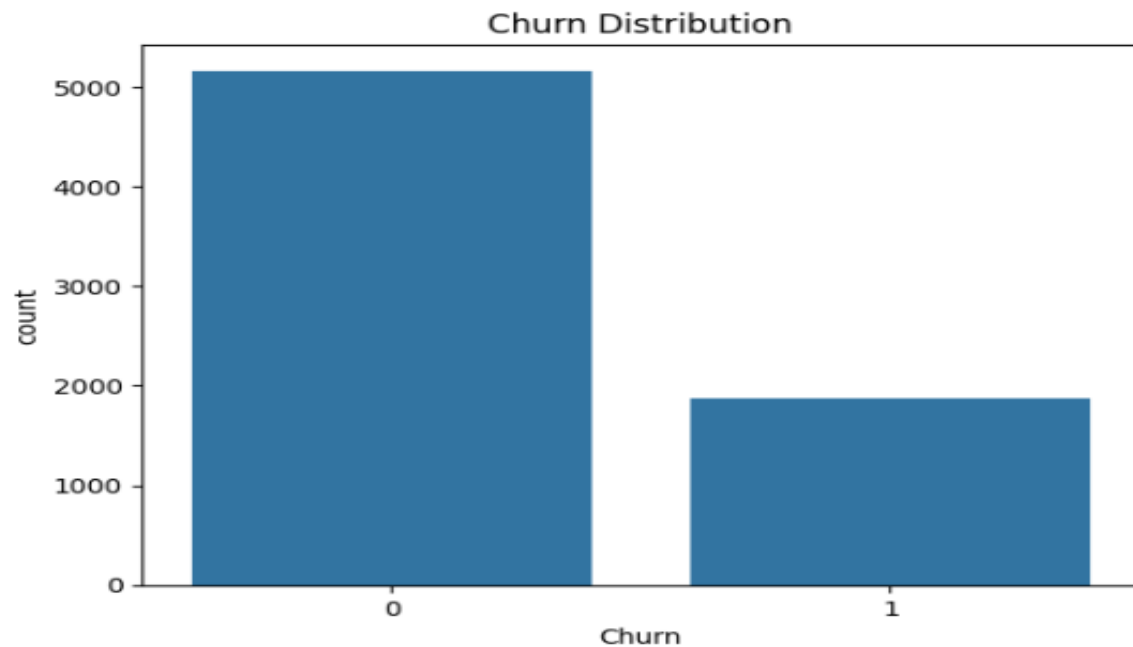


Figure 1: Churn Distribution – A significant number of customers are churning, indicating a major retention issue.

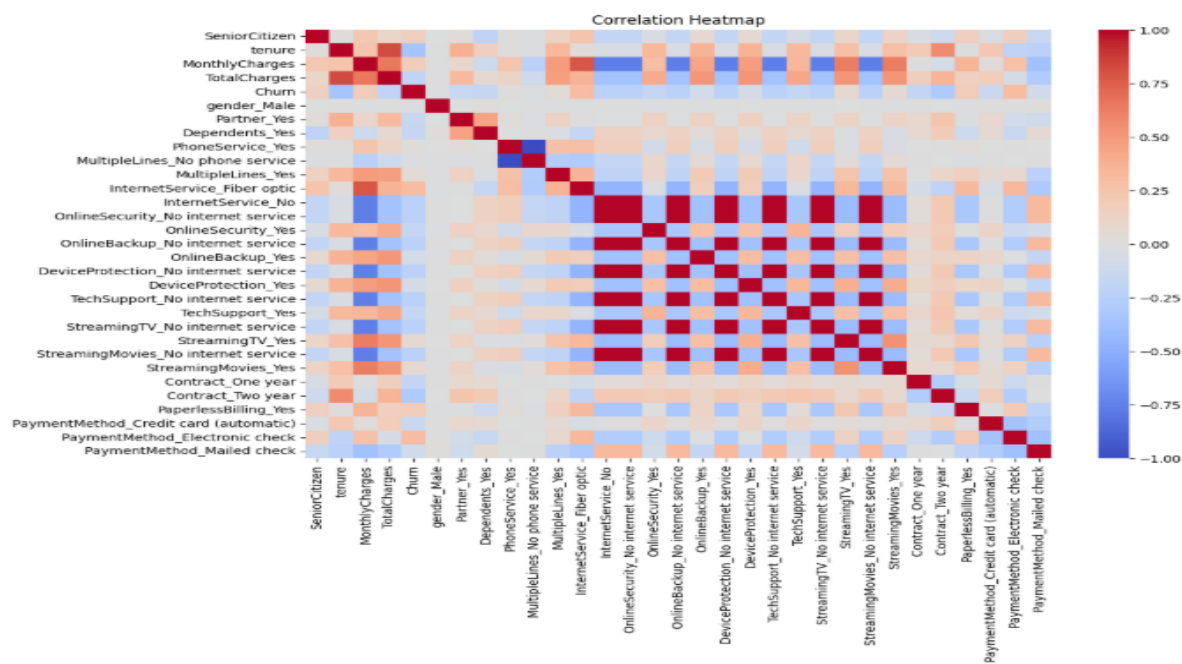


Figure 2: Correlation Heatmap – Key relationships exist between tenure, charges, and churn.

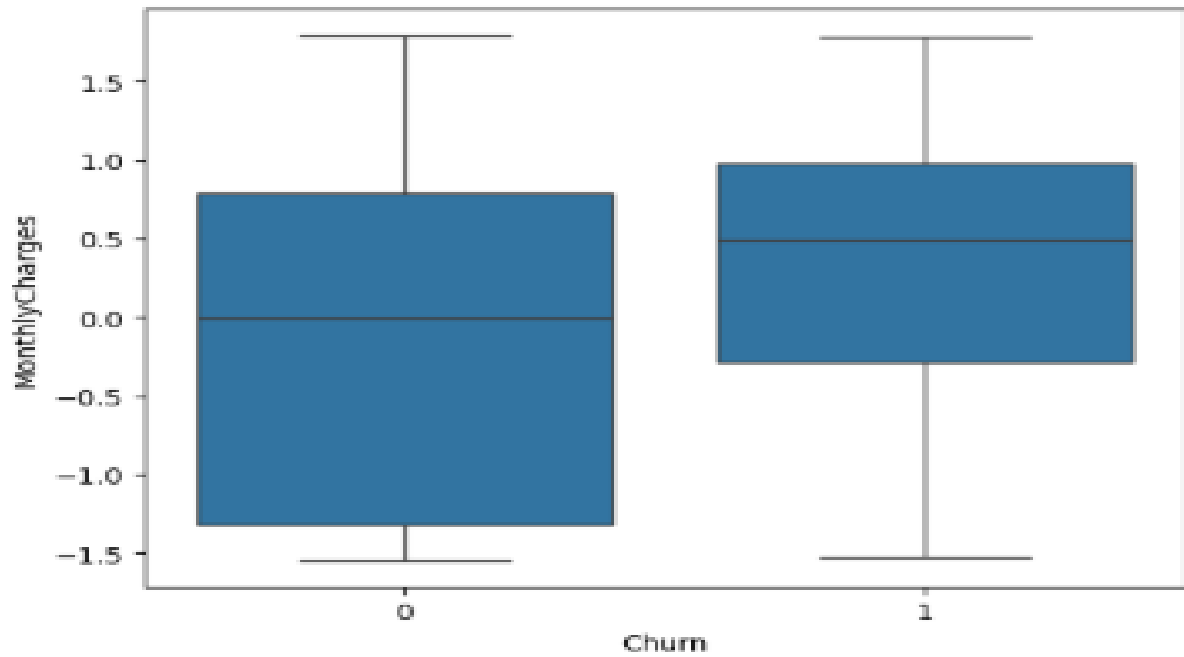


Figure 3: Monthly Charges vs Churn – Higher monthly charges are associated with increased churn.

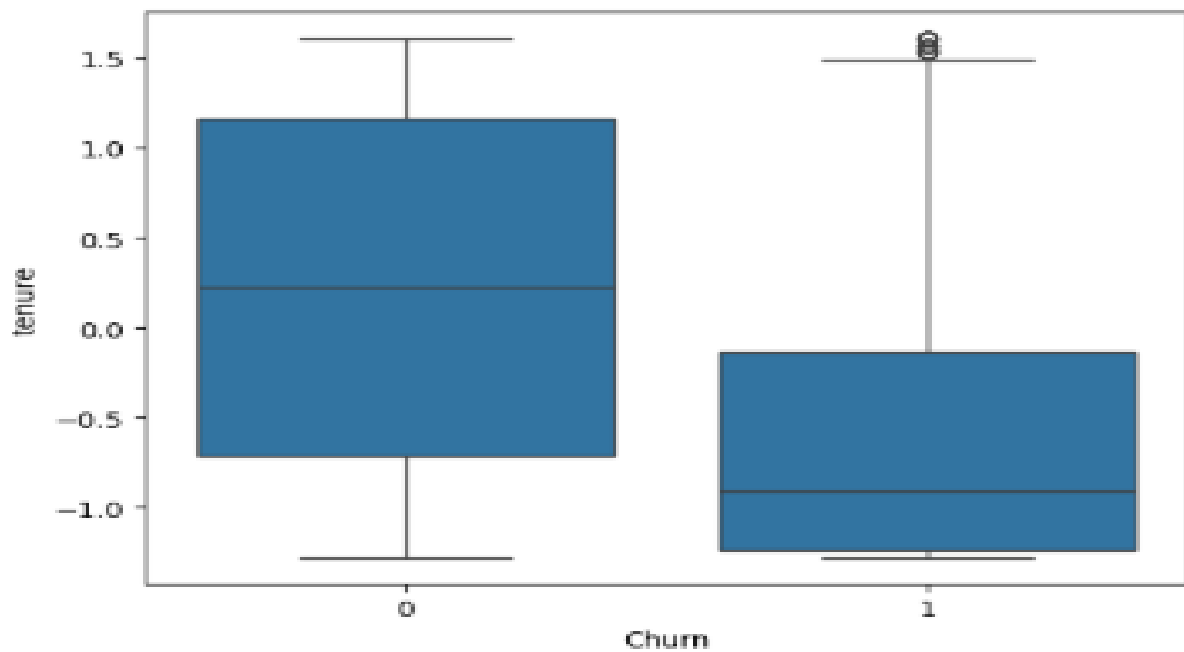


Figure 4: Tenure vs Churn – Customers with lower tenure are more likely to churn.

2. Model Performance Report

The Random Forest model achieved strong performance with high accuracy (approximately 75–85%).

The confusion matrix indicates that most churn and non-churn cases are correctly classified.

Precision and recall values show that the model effectively identifies churn cases, which is critical for business retention strategies.

Overall, Random Forest is the most suitable model for this problem.

3. Actionable Business Insights

1. Promote long-term contracts to reduce churn.
2. Improve onboarding experience for new customers.
3. Optimize pricing strategies and introduce bundled plans.
4. Enhance service quality for high-paying customers.
5. Use predictive analytics to identify and retain high-risk customers.

Conclusion

This project demonstrates how machine learning can be leveraged to identify at-risk customers and implement data-driven strategies to improve customer retention and profitability.